Petersen_R_Assignment_review_by_Valerie

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Review overall comments

- Your markdown is very cleanly written and easy to understand!
- All code ran without major error. I did get `Warning: NAs introduced by coercion` once, but the code still ran and produced the graph.

PART 1 Review

Data Inspection

- Very thorough on data inspection with each command purpose explained and answer commented.
- I liked how you described the variables important for the assignment.
- No issues, all ran smoothly.

Data Processing

- Your final files appear to be sorted alphanumerically by position and are not in increasing or decreasing order.
- Specifically, looking at `CSV_files /maize_chr_10_decrease.csv`, the positions on:
  - Line 17 (61,763,079)
  - Line 18 (6,121,326)
  - Line 19 (43,936,126)
- Line 17 is greater than Line 18, but Line 18 is less than Line 19.
- I think the error might have been when you transposed the data. You comment that the data is now character and convert to data frame, but that still left `Position` as a character.
- After your *merge* and before your *function*, I entered:

```r
typeof(Maize_merge_2$Position) # this produced Character
typeof(Teosinte_merge_2$Position) # this produced Character
```

- I think this caused your function to sort incorrectly since `Position` remains a character.
- After your *merge* and before your *function*, I added:

```r
Maize_merge_2$Position <- as.numeric(Maize_merge_2$Position)
Teosinte_merge_2$Position <- as.numeric(Teosinte_merge_2$Position)
typeof(Maize_merge_2$Position) # this produced Double
typeof(Teosinte_merge_2$Position) # this produced Double
```

- Then when I re-ran the rest of your code, the final files no longer sorted alphanumerically.
- You have `cdv_marker_id` at the end of your files and your first coloumn is titled `SAMPLE_ID`. I'd change this to fit the instructions, which say: **All our files will be formatted such that the first column is "SNP_ID", the second column is "Chromosome", the third column is "Position", and subsequent columns are genotype data from either maize or teosinte individuals.**
- Outside of that, everything else looked good and made sense.
- When I ran everything, it generated the csv files in the main folder and not the `CSV_files` folder. Not a huge deal but you could write it directly into `CSV_files` so when it's run it won't duplicate in `R_Assignment_BCB5460`
- I should have deleted all edits/test-code I added and any duplicate test-csv files generated. I only pushed my review .Rmd and .pdf so hopefully none of your original markdown code should be altered.

PART 2 Review

SNPs per chromosome

- I did it slightly different, but I think the distribution across chromosomes resulted in the same pattern!
- I like your ten plots using facet wrap. The x-axis is a bit hard to read, but that could just be my window size.

Missing data and amount of heterozygosity and homozygosity

- No issues, all ran smoothly with graph showing up.

Your own visualization

- No issues, all ran smoothly with graph showing up.

End of my review.

Valerie Yu